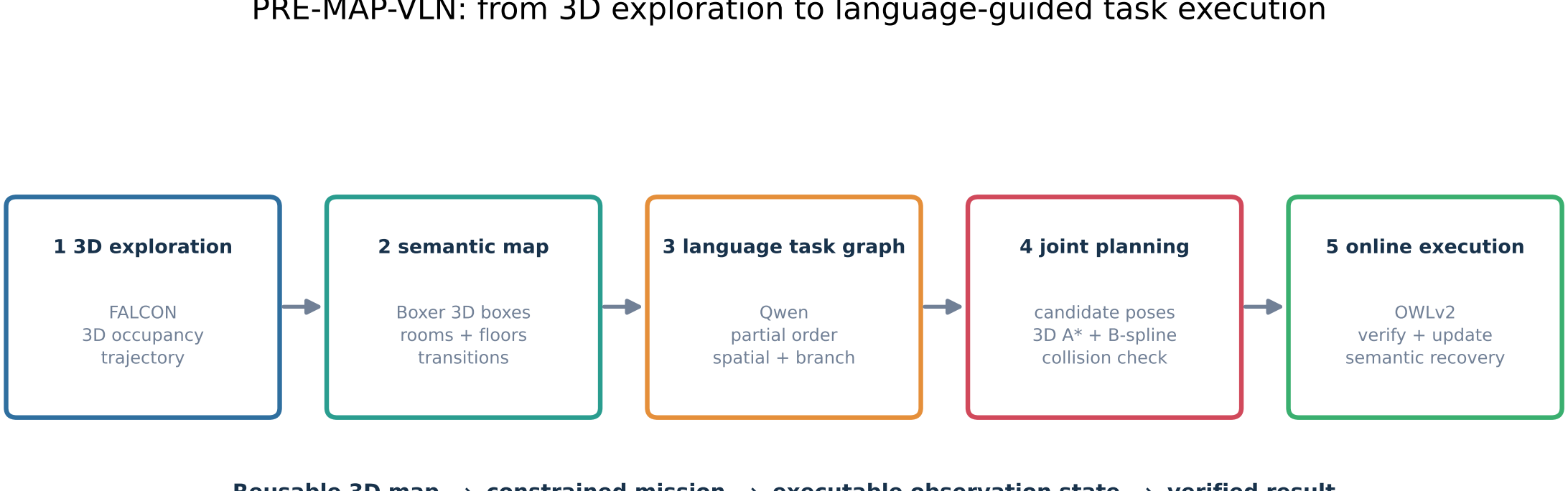
PRE-MAP-VLN: from 3D exploration to language-guided task execution
1 3D exploration
FALCON
3D occupancy
trajectory
2 semantic map
Boxer 3D boxes
rooms + floors
transitions
3 language task graph
Qwen
partial order
spatial + branch
4 joint planning
candidate poses
3D A* + B-spline
collision check
5 online execution
OWLv2
verify + update
semantic recovery
Reusable 3D map → constrained mission → executable observation state → verified result